

India's Energy Transition

500 GW Renewable Target, Nuclear Expansion, and the Coal Dilemma

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Executive Summary

India's energy transition is gathering pace, with renewable power taking centre stage as the country looks to fuel its fast-growing economy. Yet fossil fuels — coal, oil, and imported gas — remain deeply embedded in the energy mix. India has committed to achieving energy independence by 2047 and net-zero carbon emissions by 2070. The central tension is acute: India spends approximately US\$150 billion on energy imports each year, and reducing that dependence requires domestic clean-energy alternatives to reach cost-competitiveness CNA[2026-01-29]. The current trajectory of significant strides in renewable deployment, paired with continued reliance on imported Russian crude and a nascent green hydrogen programme, defines India's near-term energy reality.

Part I — Clean Energy Ambitions and Fossil Fuels in the Mix

India's clean energy transition remains firmly in the spotlight, with significant investment and infrastructure development still needed to keep the country's green plans on track CNA[2026-01-29]. The country has made meaningful strides in renewable power, and analysts recognise the transition is real — but the parallel dependence on fossil fuels has not eased quickly. India is simultaneously accelerating renewable deployment and navigating a more complex crude oil import picture driven by geopolitical disruption in the Middle East.

India's aim to become energy independent by 2047 is framed explicitly as a national security objective. The scale of the challenge is reflected in the import bill: at approximately US\$150 billion per year in energy imports, diversifying the supply base and building domestic clean generation capacity are intertwined policy imperatives CNA[2026-01-29].

China, the United States, Brazil, India, and Germany have installed significant amounts of renewable energy capacity — India among the countries that have embraced a combination of policies, including regulatory interventions and incentives, enabling such positive changes in solar and wind deployment [RAG: CLIMATE_RAG — UNEP Emissions Gap Report 2023-2024 (2023-2024)].

Part II — Crude Oil: Russia as Dominant Supplier and the Discount Calculus

Russia Becomes India's Largest Crude Supplier

Russia has become India's largest crude oil supplier since the start of the Ukraine war CNA[2025-12-04]. Before the Iran war disrupted Middle East supply chains, India imported around 2.6 million barrels per

day of crude oil from the Middle East CNA[2026-03-31]. As the conflict strained those supplies, India's crude oil imports from Russia jumped to roughly 1.9 million barrels a day in March 2026, up from approximately 1 million barrels before the Iran war began — though data suggests this Russian volume is probably not sufficient to offset the loss of Middle East supplies, particularly with peak summer energy demand approaching CNA[2026-03-31].

Reliance and the US Pressure Campaign

Reliance Industries — operator of India's Jamnagar refinery — stated it expects no Russian crude deliveries in January 2026, a day after US President Donald Trump issued a fresh warning to India to cut its imports of Russian oil or face further tariff increases CNA[2026-01-06]. Indian government sources meanwhile reaffirmed that India will continue to buy Russian oil: "These are long-term oil contracts. It is not so simple to just stop buying overnight" CNA[2025-08-03].

Shrinking Discounts Curb State Refiner Appetite

Indian refiners were pulling back from Russian crude as discounts shrank to their lowest levels since 2022, when Western sanctions were first imposed on Moscow. The compression in discount was attributed to lower Russian export volumes and steady demand. India's state refiners — Indian Oil Corp, Hindustan Petroleum Corp, Bharat Petroleum Corp, and Mangalore Refinery Petrochemical Ltd — had not sought Russian crude in the prior week, according to four sources familiar with the refiners' purchasing CNA[2025-08-03].

Asian nations more broadly are competing for Russian crude as the Iran war strains Middle East supplies. The approach of peak summer energy demand — driven by travel, agriculture, and freight needs — amplifies the pressure on Asian importers including India CNA[2026-03-31].

Part III — Green Hydrogen: Ambition, Cost Barrier, and Early Movers

The Cost Threshold

One emerging option showcased in India's clean energy transition is green hydrogen — produced by splitting water into hydrogen and oxygen using renewable electricity. It remains costly, at just under US\$4 per kilogram. India's Minister for Petroleum and Natural Gas, Hardeep Singh Puri, framed the policy trigger clearly: "If we can bring it down to below US\$3 ... there will be a public policy pressure (to) shift to green hydrogen ... since we spend US\$150 billion on imports of energy (each year)" CNA[2026-01-29].

Early Industrial Movers

JSW Energy is already making plans to build a second green hydrogen plant that will be 20 times bigger than its first CNA[2023-12-07]. CRISIL Market Intelligence and Analytics senior practice leader and Director for Energy and Sustainability Pranav Master stated that India's focus on green hydrogen would improve its energy self-reliance: the government's consultation with industry is moving in the right direction and the opportunity could become very large CNA[2023-12-07].

India is increasingly focusing on green hydrogen as part of its aim to become energy independent by 2047 and achieve net-zero carbon emissions by 2070 CNA[2023-12-07]. International partnerships are developing alongside domestic industrial investment — Prime Minister Narendra Modi's 2023 visit to Australia produced new bilateral agreements specifically on green hydrogen cooperation CNA[2023-05-25].

Funding Hurdles Remain

Significant funding hurdles stand in the way of India's green hydrogen emission targets. The domestic policy framework is moving in the right direction, but the scale of investment required to bring costs below

the US\$3/kg threshold needed to trigger widespread adoption has not yet materialised CNA[2023-12-07].

Part IV — Nuclear Energy: Civil Programme and Strategic Context

Civil Nuclear Independence

India's civil nuclear strategy has been directed toward complete independence in the nuclear fuel cycle, necessary because of its rejection of the Nuclear Non-Proliferation Treaty. Following economic and technological isolation after nuclear tests in 1974, India largely diverted its focus toward developing and perfecting fast breeder reactor technology [RAG: CONFLICT_RAG — (undated)].

Nuclear power supplied 3.1% of India's electricity in 2000, with two Russian units under consideration for further construction at Kudankulam at that time [RAG: CONFLICT_RAG — (undated)]. India's weapons material has been associated with a Canadian-designed 40 MW research reactor that began operations in 1960 and a 100 MW indigenous unit in operation since 1985, both using local uranium [RAG: CONFLICT_RAG — (undated)].

Conclusion

India's energy position in 2026 is defined by three simultaneous pressures: a clean energy transition that is real but requires sustained investment and infrastructure development to stay on track CNA[2026-01-29]; a crude oil supply picture disrupted by conflict, US sanctions pressure, and shrinking Russian discounts CNA[2025-08-03] CNA[2026-03-31]; and a green hydrogen programme that is technically promising but commercially constrained by a cost gap that must close before policy pressure translates into scaled demand CNA[2026-01-29]. India's declared goal of energy independence by 2047 and net zero by 2070 sets the strategic direction; the pace at which these three pressures are resolved will determine whether that trajectory is achievable.

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